

Energy Sector Sees Innovation Across the Value Chain

From power sources, grids and utility providers, all the way down to the end-consumers, there is a splurge of innovative technologies all across the energy sector. The current trend should hopefully lead to more reliability and sustainability in energy



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At the government, industrial and consumer levels, there is rising awareness about depleting resources, lack of reliability and sustainability in the energy sector.

"There is a growing need to manage the ever-increasing demand and peak loads in India. Load management, smart substations for medium voltage (MV) and low voltage (LV), local network reconfiguration and non-critical load shedding are some of the techniques that help us towards this end. It is also important to integrate

more green energy into our electrical grids. Plus, with the large number of sensitive electronic gadgets being used we also have to make sure that the electricity quality is up to the mark. Disturbances on the network (flickers and harmonics) impact power quality for end-users. A number of harmonic filters and automatic fault detection schemes exist that help improve the power quality. Maintaining and up-grading aging electrical assets is also a prime concern," says Dr Satish Kumar, energy efficiency ambassador and vice president, Schneider Electric India.

Understandably, a lot of new technologies are coming up to resolve these issues. On one end, there are alterna-

tive energy sources, especially solar. Then, there are smart grids and smart metering technologies that enable efficient consolidation and distribution of power. This is followed by building automation and energy management solutions, smart power outlets, energy-efficient air-conditioners and heaters, and a range of other 'smart' equipment that help monitor, manage and optimise the power consumption in homes, offices and industries.

Behind all this, there is a layer of supporting infrastructure, such as a high-speed and high-reliability communication network, and components such as capacitors, inductors and harmonic filters, insulation monitors, MV switchboards and switchgears, sensors, wireless transceivers, low-power microcontrollers, etc.

On a more integrated plane, there are solutions for building automation and energy management, tools to track carbon footprint of operations, network management technologies for utility providers, maintenance solutions that prevent electrical systems from deteriorating and becoming obsolete, and Cloud-based solutions that assist in better automation and connectivity across segments such as electric utilities, water and waste water, oil and gas, and mining and minerals.

Energy-related products are in focus for companies across the spectrum of technology. Schneider Electric, for example, has a vast product portfolio of interesting products catering to various markets—Wiser for home automation, Telvent for infrastructure management, Ecofit, Exostruxure and StruXureware for energy monitoring and control, etc.